Moving from 3D to 2D: Basic Shapes

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In this activity, students will learn how 3D objects are represented as 2D tactile graphics. The translation from 3D to 2D representations benefits from direct, explicit instruction to help students develop a strong foundation for interpreting tactile graphics.

Setup: 3D to 2D Shapes

This activity can be broken down into several elements for lesson planning.

Objectives:

- Practice exploring information in 3 dimensions: height, width, and depth
- Practice exploring information in 2 dimensions: height and width
- Match features from a 3D object to the corresponding 2D tactile graphic
- Explore how shapes can be used to create a recognizable object

Materials Needed:

- Three handheld-size models of the following shapes, approximately 2 inches in size:
  - A cube
  - A ball (sphere)
  - A 3-sided pyramid
- Three shapes, approximately 2 inches in size: A square, circle, and triangle. Choose your adventure! A teacher can prep these shapes OR this can be an activity for students to create their own shapes:
  - Using a stencil, create and cut out a square, circle, and triangle on construction paper or card stock
  - Use a stencil, and a Sensational Blackboard with copy paper. See Ann Cunningham's video *Creating Pop Out Shapes* (1:55)
  - Use the APH Geometric Drawing Stencils
- APH Draftsman or TactileDoodle
  - APH Tactile Drawing Film

- Stylus or ball point pen
- Glue stick
  - ***Teaching tip:*** Put a piece of tap or a bump dot on the glue stick to keep it from rolling off the table
- Construction paper
- Scissors

Alternate Materials:

- Any tactile drawing board, such as the Sensational Blackboard
- Plain copy paper

Ideas for carrying out this activity:

- Art or math class
- Architecture or shop class
- Extension activity when reading a book about shapes

Adventure Map: 3D to 2D Shapes

Teaching tips: Provide sufficient time for students to explore all materials before beginning the activity. Set up a work space with containers or trays to keep stylus/pens, glue stick, scissors, and drawing materials organized. It is best to use 3D objects and materials that can be sanitized or consumable. Keep hand sanitizer handy and invite students to sanitize hands before beginning the activity!

1. Introduction

- Ask your student about their experiences with the geometric shapes square, circle, triangle
  - What do the shapes remind them of? Where have they seen these shapes in day-to-day life?
- Allow time for your student to explore the various 3D models of each shape (cube, ball, pyramid)
- Discuss the features of each shape. Use descriptive language to describe the number of sides and describe if there are corners or edges to a shape. If your student needs more practice using descriptive language, review

TADA! Activity 2

- Experiment with how a stencil can be used to draw each shape on a tactile drawing board. Support troubleshooting of how to coordinate holding the stencil in place while using enough force to make tangible marks with a pen or stylus. If a student needs more practice using a tactile drawing board, review TADA! Activity 1

2. Working with shapes

- Name each geometric shape: square, circle, triangle

- Ask your student to draw a square, circle, triangle. It is important that students try each of the following methods of drawing shapes. Students can build confidence drawing shapes in all 3 ways:
  - Use a stencil
  - Trace a shape
  - Free draw (remind students that it is OK if their drawings are not perfect like the stencil - drawings do not have to be perfect to be perfectly useful!)

- Ask your student to explore what they have drawn; then ask which 2D symbol they would use to represent the sphere? The cube? The pyramid?

- Create a cutout where the space matches or is slightly larger than the size of the 3D object. Have the student push the object through the cutout to see how the negative space matches the 3D object

- Discuss: When a 3D object is depicted as a 2D picture, it has to be distorted because we have eliminated depth. But, a 3D object can still be sufficiently represented by a 2D symbol/graphic

3. The importance of outlines

- An outline is important when drawing a recognizable shape, but it doesn't have to be perfect to be useful.
  - For example: a hand is a very complex shape, yet you can break it down into geometric shapes and create a recognizable object

- Using a tactile drawing board, have your student draw an outline of their hand

- Next, have your student explore their own hand. You can say:
  - Starting with your palm, which shape would you use to represent the

palm?

    * Using the construction paper, cut a shape out to represent the palm. Select a new piece of paper and glue it on while keeping enough space to glue your thumb on later. Just use a little bit of glue for now to tack it down on the paper - you can move it around later if needed

  - Now point to your finger. Which shape would you use to represent the finger?
    * Cut out a shape for each finger. Where does each finger attach to the palm? (All 4 fingers need to be on the same side of palm)

  - What's the difference between the shapes of different fingers?

  - Now let's look at the thumb - it is not a basic shape!
    * When you are ready to attach it to the hand, look at where it attaches to the palm

  - Wow! You have made a really complex shape! Yay!

  - ***Teaching tip:*** Introduce the correct vocabulary to identify the different shapes (rectangle, oval, circle, etc.) AND the parts of the hand (palm, fingers, thumb)

4. **Sharing and Discussion**

- Have each student share and appreciate each other's **hand**iwork
  - Encourage students to identify the shapes that they discover on one another's newly constructed hands

  - Identify similarities and differences between one another's hands

- Encourage the students to discuss any surprises or troubleshooting they encountered

- Guide discussion on how construction of their hand did not have to be perfect in order for it to be useful in representing a hand

5. **Conclusion**

- Summarize the shapes that were explored and which 2D symbols were used to represent which 3D objects
  - Circle = ball or sphere

  - Square = cube

- Triangle = 3-sided pyramid

- Brainstorm
  - What other shapes might be in their immediate environment? These shapes could be used to create other drawings. Have students share these shapes with one another and encourage students to create/share shapes for shared exploration. *Note:* Some students might have gaps in their experiential learning and need hands-on exploration of familiar and unknown shapes
    * Rectangles
    * Stars
    * Teardrop
  - How might shapes be used to recreate other familiar objects?
  - Frequently, shapes are used as icons or symbols. Examples:
    * A Nike swoosh
    * A religious cross
    * Star of David
    * A sun
    * 5-pointed star (can be a star in the sky or a starfish or one state on our flag)

Extension Activities:

- Draw a leaf or lots of different types of leaves
- Draw a favorite toy
- Draw a person from an articulated model (see photo below). Tracing is very difficult because the model moves too easily; instead, tape it down OR analyze what the shape of each segment is and draw the segments in proper orientation or cut them out and reassemble them into a proper relationship to each other.

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An articulated model is created from cardboard shapes that resemble each limb (forearm, upper arm, neck, torso, head, hips, thighs, and lower leg with foot. Each limb is connected at the joint with a brass paper fastener

Figure 1: An articulated model is created from cardboard shapes that resemble each limb (forearm, upper arm, neck, torso, head, hips, thighs, and lower leg with foot. Each limb is connected at the joint with a brass paper fastener